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import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df =
pd.read_csv('unemployment_data.csv')
df['Date'] =
pd.to_datetime(df['Date'])

sns.set_style('whitegrid')

# Chart 1: National average
unemployment rate over time (shows
the Covid spike clearly)
national_trend = df.groupby('Date')
['Estimated Unemployment Rate
(%)'].mean().reset_index()

plt.figure(figsize=(11, 5))
plt.plot(national_trend['Date'],
national_trend['Estimated
Unemployment Rate (%)'],
         marker='o', linewidth=2,
         color='#d62728')
plt.axvspan(pd.Timestamp('2020-03-
01'), pd.Timestamp('2020-06-30'),
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        alpha=0.2, color='red',
        label='Covid-19 Lockdown Period')
plt.title('National Average
Unemployment Rate Over Time',
fontsize=14, fontweight='bold')
plt.xlabel('Date')
plt.ylabel('Unemployment Rate (%)')
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('chart1_national_trend.png', dpi=120)
plt.close()
print("Saved
chart1_national_trend.png")

# Chart 2: Rural vs Urban comparison
over time
area_trend = df.groupby(['Date',
'Area'])['Estimated Unemployment Rate
(%)'].mean().reset_index()

plt.figure(figsize=(11, 5))
sns.lineplot(data=area_trend,
x='Date', y='Estimated Unemployment
Rate (%)', hue='Area', marker='o')
plt.axvspan(pd.Timestamp('2020-03-
01'), pd.Timestamp('2020-06-30'),
alpha=0.15, color='red')

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plt.title('Rural vs Urban
Unemployment Rate Over Time',
fontsize=14, fontweight='bold')
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('chart2_rural_vs_urban.png', dpi=120)
plt.close()
print("Saved
chart2_rural_vs_urban.png")

# Chart 3: Top 10 most affected
states during peak covid period (Apr-
Jun 2020)
covid_period = df[(df['Date'] >=
'2020-04-01') & (df['Date'] <= '2020-
06-30')]
state_impact =
covid_period.groupby('Region')
['Estimated Unemployment Rate
(%)'].mean().sort_values(ascending=False).head(10)

plt.figure(figsize=(10, 6))
state_impact.plot(kind='barh',
color='#ff7f0e')
plt.title('Top 10 States: Average
Unemployment Rate (Apr-Jun 2020)',
fontsize=14, fontweight='bold')

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plt.xlabel('Unemployment Rate (%)')
plt.gca().invert_yaxis()
plt.tight_layout()
plt.savefig('chart3_top_states_covid.
png', dpi=120)
plt.close()
print("Saved
chart3_top_states_covid.png")

# Print key numbers for the report
pre_covid = df[df['Date'] < '2020-03-
01']['Estimated Unemployment Rate
(%)'].mean()
during_covid = df[(df['Date'] >=
'2020-04-01') & (df['Date'] <= '2020-
06-30')]['Estimated Unemployment Rate
(%)'].mean()
post_covid = df[df['Date'] > '2020-
06-30']['Estimated Unemployment Rate
(%)'].mean()

print(f"\nPre-Covid average
unemployment rate: {pre_covid:.2f}%")
print(f"During-Covid (Apr-Jun 2020)
average: {during_covid:.2f}%")
print(f"Post-Covid (Jul-Oct 2020)
average: {post_covid:.2f}%")
print(f"Increase during lockdown: +

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{during_covid - pre_covid:.2f}  
percentage points")
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